

L2.2 The Mean Value Theorem

Work in your group. Check your answers against the Quarto tonight.

Rolle's Theorem

1. Verify that the function satisfies the three hypotheses of Rolle's Theorem on the given interval. Then find all numbers c that satisfy the conclusion.

$$f(x) = \sin\left(\frac{x}{2}\right) \quad \text{on} \quad \left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$$

Stop. Name the theorem, then check its hypotheses one at a time, and only then use the conclusion.

2. Rolle's Theorem does not apply to three of these. For each one, say whether it applies, and if it does not, name **which hypothesis fails**.

(a) $f(x) = x^2 - 6x + 8$ on $[1, 5]$

(b) $f(x) = |x - 2|$ on $[0, 4]$

(c) $f(x) = \frac{1}{x-1}$ on $[0, 2]$

(d) $f(x) = x^2$ on $[0, 3]$

3. Let $f(x) = \tan x$. Then $f(0) = f(\pi)$, but there is no number c in $(0, \pi)$ with $f'(c) = 0$. **Why does this not contradict Rolle's Theorem?**

The Mean Value Theorem

4. Verify that the function satisfies the hypotheses of the Mean Value Theorem on the given interval. Then find all numbers c that satisfy the conclusion.

$$f(x) = 2x^2 - 3x + 1 \quad \text{on } [0, 2]$$

5. Same instructions, and this one needs its domain checked before anything else.

$$f(x) = \frac{1}{x} \quad \text{on } [1, 3]$$

6. $f(x) = x^3 - 3x^2$ on $[0, 3]$. Find every c that satisfies the conclusion of the Mean Value Theorem — and say which candidates you had to **reject**, and why.

Stop. Two things to notice before you finish: what m_{sec} came out to, and which interval the conclusion lives on.

7. Let $f(x) = 2 - |2x - 1|$. Show that there is no value of c with $f(3) - f(0) = f'(c)(3 - 0)$. **Why does this not contradict the Mean Value Theorem?**

Exactly one real root

8. Show that the equation $2x - 1 - \sin x = 0$ has **exactly one** real root.

Stop. Two claims, two different theorems. Which one gives *at least one*, and which one rules out a second?

9. Suppose f is differentiable everywhere.

(a) Suppose f has two roots. Show that f' has at least one root.

(b) Suppose f is twice differentiable and has three roots. Show that f'' has at least one real root.

Functions with the same derivative

10. If $f'(x) = c$ for all x , where c is a constant, use Corollary 7 to show that $f(x) = cx + d$ for some constant d .

11. Let $f(x) = \frac{1}{x}$ and

$$g(x) = \begin{cases} \frac{1}{x}, & x > 0 \\ 1 + \frac{1}{x}, & x < 0 \end{cases}$$

Show that $f'(x) = g'(x)$ for every x in their domains. Can you conclude from Corollary 7 that $f - g$ is constant?

Stop. Read the hypothesis of Corollary 7 again, one word at a time. What is the domain of f , written as intervals?

Lawyer Math

12. **BOARDS** **Group notes.** For each theorem below: write the statement **exactly**, list its hypotheses one per line, and then do the problem beside it.
- ⑤ **Rolle's Theorem.** Verify that $f(x) = x^3 - 2x^2 - 4x + 2$ satisfies the three hypotheses on $[-2, 2]$, then find all numbers c that satisfy the conclusion.
 - ⑥ **Mean Value Theorem.** Verify that $f(x) = x^3 - 3x + 2$ satisfies the hypotheses on $[-2, 2]$, then find all numbers c that satisfy the conclusion.

With these two the set of six is complete. **On your board, put all six in order** and mark each one as *existence*, *location*, or *procedure*.

① IVT ② EVT ③ Fermat ④ Closed Interval Method ⑤ Rolle ⑥ MVT